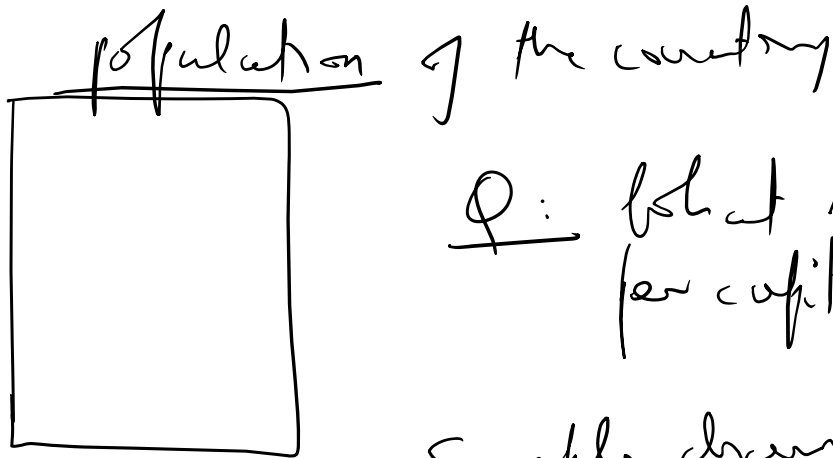


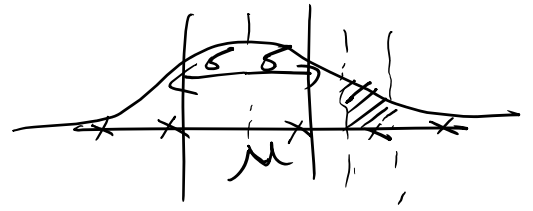


Q: What is the average per capita income?

Sample drawn from a rich neighbourhood in Mumbai is highly unrepresentative.

Sample must be drawn in accordance to
class numbers $\rightarrow$ requires a
scientific method.

Suppose we have a population and we
draw n samples out of the population



Sampling with replacement

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \quad \left[\text{sample mean} \right]$$

a random variable? Each of the X_i 's is a

random variable.

$$E(X_i) = \underline{\mu}$$

$$\text{Var}(X_i) = \underline{\sigma^2}$$

What is the value of $E(\bar{X})$?

Linearity of expectation means

$$E(\bar{X}) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right)$$

$$= \frac{E(X_1) + E(X_2) + \dots + E(X_n)}{n}$$

$$\frac{\mu + \mu + \dots + \mu}{n} = \frac{n\mu}{n} = \underline{\mu}$$

$$\text{Var}(\bar{X})$$

$$= \text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = ?$$

$$\text{Var}(kX) = k^2 \text{Var}(X) \quad \text{property}$$

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) \quad \text{when } X_1 \text{ and } X_2$$

are independent

$$\text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \text{Var}\left(\frac{X_1}{n}\right) + \text{Var}\left(\frac{X_2}{n}\right) + \dots + \text{Var}\left(\frac{X_n}{n}\right)$$

because $X_1, X_2, \dots, X_n$ are independent

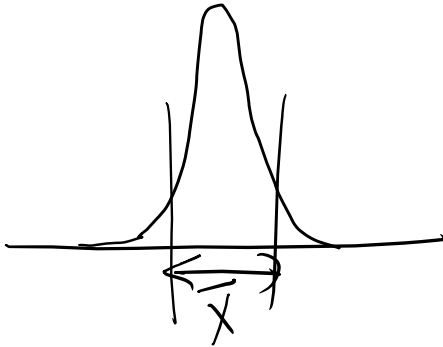
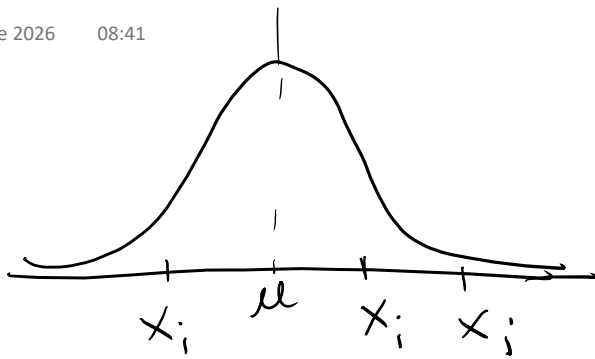
$$\text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right)$$

$$= \frac{1}{n^2} \text{Var}(X_1) + \frac{1}{n^2} \text{Var}(X_2) + \dots + \frac{1}{n^2} \text{Var}(X_n)$$

$$\frac{\sigma^2}{n^2} + \frac{\sigma^2}{n^2} + \dots + \frac{\sigma^2}{n^2} = \frac{n\sigma^2}{n^2}$$

$$= \frac{\sigma^2}{n}$$

$$\boxed{\text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}}$$



$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Extreme values on one side will be cancelled by extreme values on the other side
 → leading to lower variance of $\bar{X}$

We computed mean & variance of $\bar{X}$

We can say more $\rightarrow$ What is the distribution of $\bar{X}$?

The Central Limit Theorem says that

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Probability sampling

population

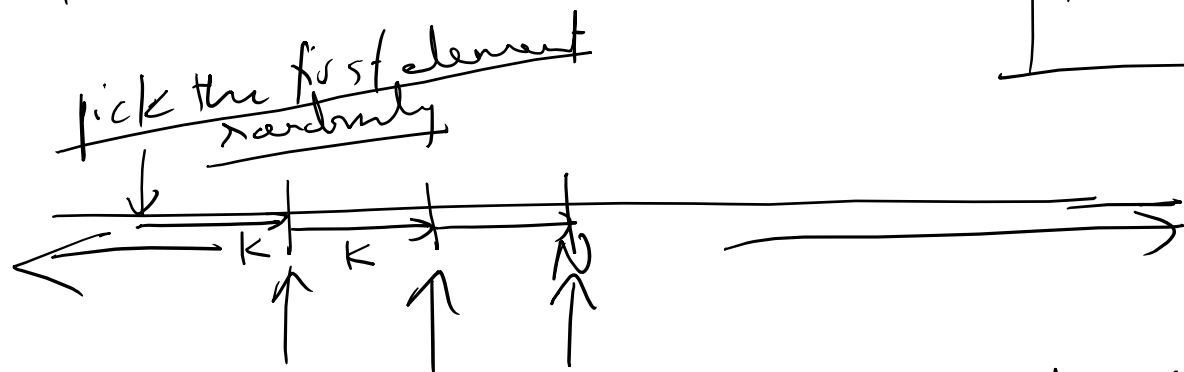
1	2	...	N
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Label the entire population from 1 to N.
draw a uniform random sample from

1 to N.

Systematic random sampling

$$K = \frac{N}{n}$$



remaining samples at regular intervals



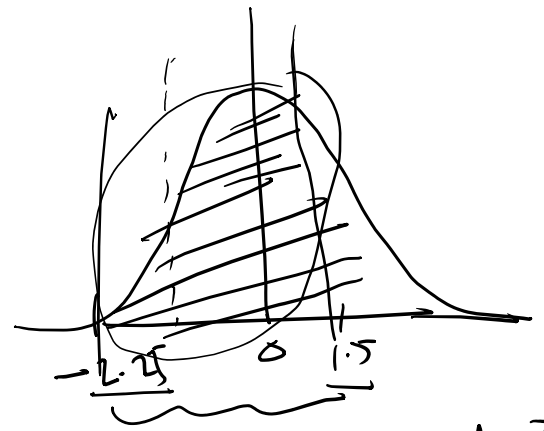
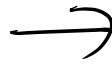
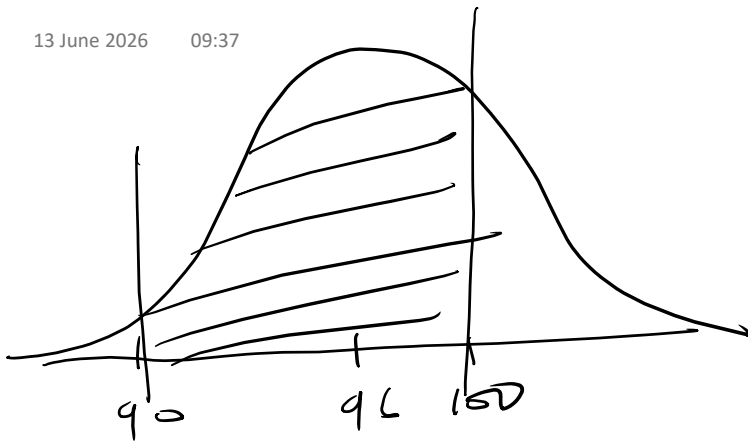
standard normal
distribution for
the population

sample mean
= -1

and the sample size = 100.

The distribution of $Y = \frac{X_1 + X_2 + \dots + X_n}{n}$ is normally distributed when the X_i s are normally distributed for any n

When $X_1, X_2, \dots, X_n$ follow any distribution, then $\frac{X_1 + X_2 + \dots + X_n}{n}$ will have a normal distribution as $n \rightarrow \infty$



easy to compute
from a table

$$P(Z \leq 1.5) - P(Z \leq -2.25)$$

In case of infinite populations, there is no
difference between sampling with
replacement and sampling without
replacement

In case of finite populations, there is a
difference

To calculate sample proportion

$$Y = \frac{X_1 + X_2 + \dots + X_n}{n} \quad (\text{just like sample mean})$$

but each X_i is a 0 or 1 random variable where $X_i = 0$ if the i th sample does not have the characteristic and $X_i = 1$ otherwise

$$E(Y) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right)$$

$$= \sum E\left(\frac{X_i}{n}\right) \text{ by linearity of expectation}$$

$$E\left(\frac{X_i}{n}\right) = \frac{1}{n} \underbrace{E(X_i)}_{\text{Bernoulli random variable}}$$

$$E(X_i) = P(X_i = 1) \cdot 1 + P(X_i = 0) \cdot 0$$

$$E(X) = \sum i P(X = i)$$

$$\rightarrow E(X_i) = P$$

$$E\left(\frac{X_i}{n}\right) = \frac{P}{n}$$

$$E(Y) = \sum E\left(\frac{X_i}{n}\right) = \cancel{n} \cdot \frac{P}{\cancel{n}} = \underline{P}$$

$$\text{Var}(Y) = \text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right)$$

$$= \frac{1}{n^2} \times n \text{Var}(X_i)$$

$$= \frac{\text{Var}(X_i)}{n}$$

→ Bernoulli
random
variable

$$\boxed{\text{Var}(Y) = \frac{p(1-p)}{n}}$$

Confidence interval

Suppose we have a population with an unknown parameter θ

Collect a random sample $X_1, \dots, X_n$ and compute an estimate

$$\hat{\theta} = g(X_1, X_2, \dots, X_n)$$

Example 1 $g = \frac{X_1 + X_2 + \dots + X_n}{n}$

The estimator has a probability distribution
called its sampling dist

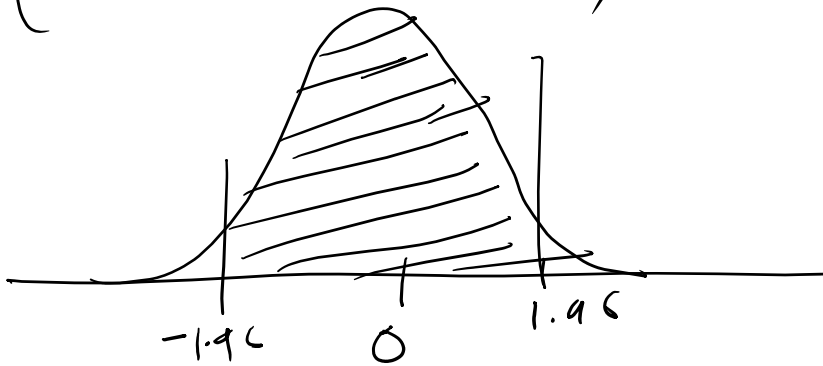
$$X_i \sim N(\mu, \sigma^2)$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ or } \underline{\text{equivalently}}$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad [\text{Z-score}]$$

The standard normal distribution tells us

$$P(-1.96 < Z < 1.96) = 0.95$$



$$P\left(-1.96 < \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < 1.96\right) = 0.95$$

$$\boxed{\frac{-1.96\sigma}{\sqrt{n}} < \bar{X} - \mu < \frac{1.96\sigma}{\sqrt{n}}}$$

$$\mu > \bar{X} - \frac{1.96\sigma}{\sqrt{n}}$$

$$\mu < \bar{X} + \frac{1.96\sigma}{\sqrt{n}}$$

μ lies in the interval $\left[\bar{X} - \frac{1.96\sigma}{\sqrt{n}}, \bar{X} + \frac{1.96\sigma}{\sqrt{n}}\right]$
with probability 0.95

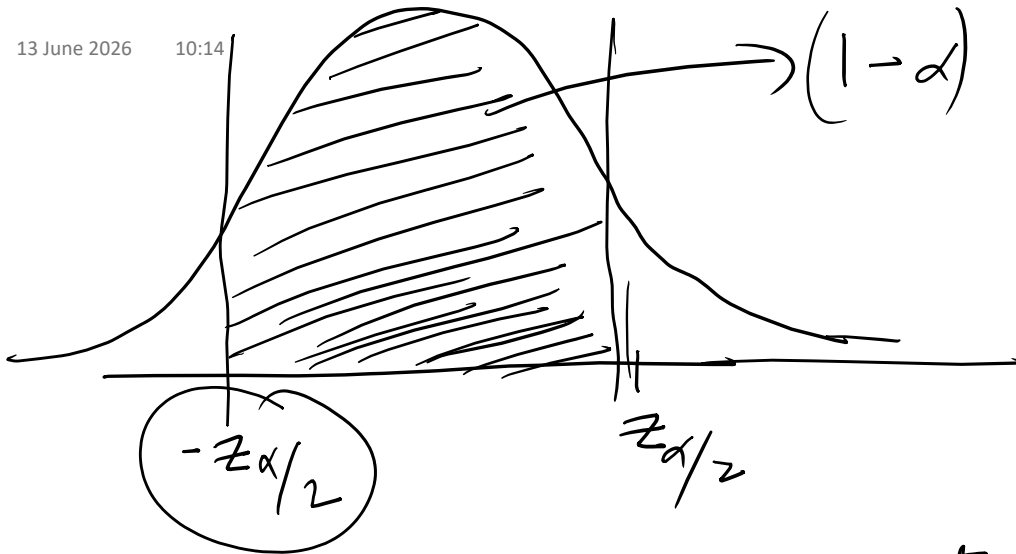
$\bar{X} \pm \frac{1.96\sigma}{\sqrt{n}}$ is a 95% confidence interval for μ

General form

$$P(-c < Z < c) = 1 - \alpha$$

$(1 - \alpha)$ is the confidence interval and c is the critical value

For a normal distribution $c = Z_{\alpha/2}$



What is the interpretation of the confidence interval?

$$\bar{X} - \frac{1.96\sigma}{\sqrt{n}} < \mu < \bar{X} + \frac{1.96\sigma}{\sqrt{n}}$$

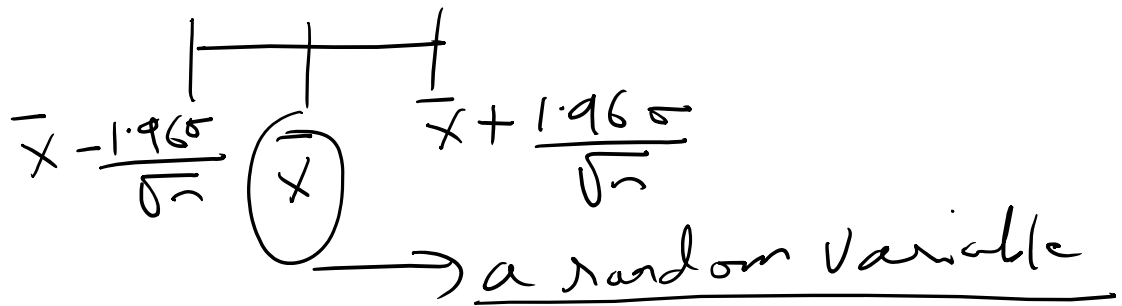
with probability 0.95

Bayesian interpretation

is that μ is a random variable
and with prob 0.95, it lies between
 $\bar{X} - \frac{1.96\sigma}{\sqrt{n}}$ and $\bar{X} + \frac{1.96\sigma}{\sqrt{n}}$

frequentist interpretation

→ μ is not a random variable, it is a fixed constant which is unknown



95% of the time, the interval
 $\bar{x} - \frac{1.96\sigma}{\sqrt{n}}$ and $\bar{x} + \frac{1.96\sigma}{\sqrt{n}}$
contains μ